

Roberto Reyes

reyes.rrrob@gmail.com | (760)-877-1834 | roreye5.github.io | github.com/roreye5

EDUCATION

University of California, San Diego

June 2025

B.S. Electrical Engineering - Computer System Design Depth

Relevant Coursework: Embedded Systems Design & Interfacing, Components & Circuits Laboratory, Digital System Design, Product Engineering & IoT Development, Python for Data Analysis, Fast Prototyping & Mechatronic Systems, Object Oriented Programming

TECHNICAL SKILLS

Programming: C, C++, Python, MATLAB

Hardware: PCB Assembly & Bring-Up, SMD/Through-Hole Soldering, Circuit Debugging, Hardware Validation, Oscilloscope Measurements, Multimeter Measurements, Power Supply Testing, Schematic Review, KiCad

Embedded Platforms: STM32 (STM32CubeIDE), ESP32 (ESP-IDF), Arduino

Interfaces & Protocols: UART, I2C, SPI, GPIO, Bluetooth, ESP-NOW, TCP/IP

Tools & Concepts: FreeRTOS, Git, LTSpice, debugging, multithreading fundamentals, system integration, SolidWorks, Onshape

EMBEDDED SYSTEMS PROJECTS

Embedded Decentralized Wireless Communication Device *(C++, Embedded Systems, Wireless Protocol Development)*

Designed and built a portable wireless messaging system operating without cellular or Wi-Fi infrastructure.

- Developed embedded C++ firmware on ESP32 implementing ESP-NOW communication, enabling reliable device-to-device messaging without external infrastructure
- Designed and implemented a custom wireless communication protocol with packet framing, parsing, message relaying, and duplicate message detection within a 250 byte payload constraint, enabling multi-node data transfer
- Performed field testing to evaluate signal range, packet loss, and network reliability, documenting results that guided design improvements and extended communication distance to approximately 400 m through message relaying
- Executed PCB bring-up by soldering SMD components and using multimeter and bench power supplies to verify hardware functionality and diagnose integration issues

Embedded Smartwatch Sensor System *(C++, Python, Embedded Systems, I2C, Bluetooth)*

Developed a prototype device featuring pedometer, heart-rate monitoring, idle detection, and real time data display.

- Implemented real time sensor acquisition firmware using I2C peripherals with deterministic sampling, ensuring reliable data collection across multiple sensors
- Developed embedded firmware and Python data processing pipeline to acquire, process, and validate sensor data over BLE and UART, enabling real time analysis and display of sensor outputs
- Designed synchronization and communication logic between MCU and host system, ensuring stable Bluetooth data transfer and correct data alignment
- Validated sensor performance through iterative testing and calibration, improving pedometer accuracy to ~90% and ensuring consistent system behavior

Electromechanical Pinball Control System *(C++, Embedded Control Systems, Hardware Integration)*

Designed and built a working pinball machine integrating mechanical, electrical, and firmware subsystems.

- Implemented embedded firmware in C++ using a state machine to manage sensors, actuators, and system behavior, enabling real time response to user inputs
- Integrated motors, sensors, and power electronics into a unified system, supporting coordinated hardware and software interaction
- Defined system requirements and validated subsystem behavior through iterative integration and testing
- Debugged hardware and firmware issues using oscilloscopes, multimeters, and signal generators, isolating faults across system components and improving stability

Electrical Engineering Laboratory Experience – UC San Diego *(Analog and Digital Circuit Design, Testing, and Validation)*

- Built and tested analog, digital, and mixed-signal circuits using breadboards and engineering schematics, validating performance against theoretical and simulated expectations
- Used oscilloscopes, signal generators, multimeters, and power supplies to measure circuit behavior and collect test data
- Diagnosed instability, grounding issues, and faulty components through structured troubleshooting, improving circuit reliability and measurement accuracy
- Documented measurement results and test observations to produce repeatable validation procedures and clear technical reports